

Multiple Choice (10 marks)

1. Ten grams of sugar is dissolved in 100 grams (g) of water. How many grams is the sugar-and-water solution?
 - (a) 90 g
 - (b) 100 g
 - (c) 110 g
 - (d) 1000 g
2. The term marketing refers to:
 - (a) new product concepts and improvements.
 - (b) advertising and promotion activities.
 - (c) a philosophy that stresses customer value and satisfaction.
 - (d) planning sales campaigns.
3. In any ecosystem, organisms interact with biotic and abiotic parts of the environment. As a result of all of the interactions, the matter within the system is
 - (a) produced when plants grow.
 - (b) recycled when organisms die.
 - (c) lost when animals and plants die.
 - (d) formed when new animals are born.
4. Sailors guided themselves with the night sky using a
 - (a) the sun
 - (b) the moon
 - (c) never moving star
 - (d) light houses
5. Which of the following best explains what will occur when a population grows larger than the carrying capacity of its environment?
 - (a) The emigration rate will decrease.
 - (b) The competition rate will decrease.
 - (c) The birth rate will increase.
 - (d) The death rate will increase.

True / False (10 marks)

Answer True or False and justify briefly.

1. True or False: The correct answer to 'Which factor would have the greatest influence on the southern migration of animals living in the Northern Hemisphere?' is 'increasing daylight hours'.
2. True or False: The correct answer to 'Which of the following is not a transport layer vulnerability?' is 'Unauthorized network access'.
3. True or False: The correct answer to 'Loss of the ability to understand language results from loss of tissue in which of the following lobes?' is 'right parietal'.
4. True or False: The correct answer to 'Which function do kidneys provide to maintain homeostasis?' is 'the removal of carbon dioxide from the body'.
5. True or False: The correct answer to 'Our only star provides us with energy that is' is 'reusable'.

Long Form Response (20 marks)

Provide thorough reasoning and reference key steps.

1. Write a function to generate all sublists of a given list.
2. Write a function to create a list containing the power of said number in bases raised to the corresponding number in the index using map function.

End of Paper